Benjamin Haake | Curriculum Vitae

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Education

PhD in Mathematics — **Edinburgh**
University of Edinburgh, supervised by Tudor Dimofte — *2022–2026*
Research on topological quantum field theories and generalised symmetries. See below for publications.

Master of Science in Physics — **Heidelberg**
Heidelberg University — *2018–2021*
Master's thesis: Higher Gauge Theory and Its Applications, supervised by Prof. Johannes Walcher.

Bachelor of Science in Physics — **Heidelberg**
Heidelberg University — *2015–2018*
Bachelor's thesis: General Relativity in the Framework of Principal Bundles and Einstein–Cartan Theory as a Gauge Theory, supervised by Prof. Matthias Bartelmann.

Publications

Generalised symmetries, orbifold data, and gauging
PhD thesis, University of Edinburgh, doi:10.7488/era/7865 — *Oct 2026*
This thesis comprises the following two papers, a more comprehensive exposition, and the classification of gaugeable 0-form symmetries in 3-dimensional Reshetikhin–Turaev theories, as well as several smaller results.

2-Group Symmetries of 3-dimensional Defect TQFTs and Their Gauging
arXiv:2506.08178 with Nils Carqueville — *9 Jun 2025*
We study 2-group symmetries of 3-dimensional TQFTs. We explain that these symmetries can be gauged to produce new TQFTs iff certain defects satisfy the axioms of orbifold data. In the special case of Reshetikhin–Turaev theories coming from G-crossed braided fusion categories $\mathcal{C}_G^\times$, we show that there are 0- and 1-form symmetries which have no obstructions to gauging. We prove that gauging the 0-form G-symmetry on the neutral component $\mathcal{C}_e$ of $\mathcal{C}_G^\times$ produces its equivariantisation $(\mathcal{C}_G^\times)^G$, which in turn features a generalised symmetry whose gauging recovers $\mathcal{C}_e$.

Gauging 0-form, 1-form and 2-group symmetries in Reshetikhin–Turaev theories
arXiv:2605.16942 — *May 2026*
We develop additional tools and connect the construction of the previous paper to zesting. We calculate various examples of gauging that are of interest in physics, such as symmetries of Tambara–Yamagami categories (with Ising models as special cases), 1-form and 2-group symmetries in (Dijkgraaf–Witten) gauge theories, and reproduce level-dependent anomalies in Chern–Simons theory.

Talks

Gauging Higher-Group Symmetries — **Glasgow**
AGQ Centre for Doctoral Training, AGQ Student Conference — *4 Nov 2025*

Gauging 2-group Symmetries in 3d via Orbifold Data — **Online**
Beijing Institute of Mathematical Sciences and Applications, — *20 May 2025*
TQFT and Higher Symmetries Seminar; abstract and video recording available online

Higher-form Symmetries and How to Gauge Them — **Vienna**
University of Vienna, Mathematical Physics Seminar — *8 Apr 2025*

Koszul duality and the Deligne category — **Edinburgh**
University of Edinburgh, GRIFT Seminar — *18 & 25 Feb 2025*

Geometry of the BV-formalism: Supermanifolds and Symplectic Geometry — **Edinburgh**
University of Edinburgh, GRIFT Seminar — *9 Apr 2024*

From Landau–Ginzburg to Matrix factorisations and TQFT — **Edinburgh**
University of Edinburgh, STAMP Seminar — *11 May 2023*
Abstract available online

Introduction to Generalised Symmetries — **Heidelberg**
Heidelberg University, Mathematical Physics Seminar — *8 Jun 2023*

2-groups and Internalisation — **Glasgow**
Glasgow-Maxwell School, GlaMS Example Showcase — *Mar 2023*

Dijkgraaf–Witten theory — **Edinburgh**
University of Edinburgh, GRIFT Seminar — *7 & 14 Nov 2022*

Teaching Experience

Course Instructor — **Windischleuba**
Winter Academy of the CdE e.V. — *28 Dec 2025 – 5 Jan 2026*
Ran an introductory course on *basic algebra and group theory* and an advanced course on *representation theory of finite groups and category theory* (about 20 hours each), aimed at students who had no background in mathematics or who were studying at an undergraduate level.

Seminar Organiser — **Edinburgh**
University of Edinburgh — *Winter Term 2023 – Winter Term 2024*
Organised the *Fox & Hedgehog Seminar*, which gives early postgraduate students insight into big-picture problems in fields at the intersection of mathematics and physics.

Tutor — **Edinburgh**
University of Edinburgh — *Since Summer Term 2023*
Tutoring for the undergraduate courses *Fundamentals of Pure Mathematics* (ST 2023, 2024), *Introduction to Linear Algebra* (WT 2023, 2025), *Proofs and Problem Solving* (ST 2025), and *Honours Analysis Presentation Skills* (WT 2023), and for the drop-in study space *MathsBase* (ST 2024, WT 2024).

Course Instructor — **Oberwesel**
Winter Academy of the CdE e.V. — *28 Dec 2018 – 5 Jan 2019*
Ran a course on *differential geometry* (about 40 hours) aimed at students who had just begun university-level studies. Content included the basics of differentiable manifolds, vector bundles, vector and tensor fields, and Lie groups.

Tutor — **Heidelberg**
Heidelberg University — *Summer Terms 2018–2020*
Tutored for the courses *Theoretical Physics 2*, *General Relativity*, and *Advanced Mathematics for Physicists 2*.

Schools and Conferences

Workshop: Maths, Physics, and AI Alignment — **Edinburgh**
AGQ Centre for Doctoral Training — *11–12 May 2026*

AGQ Student Conference — **Glasgow**
AGQ Centre for Doctoral Training — *4–5 Nov 2025*

The Physics and Mathematics of Boundaries, Impurities, and Defects — **Cambridge**
Isaac Newton Institute for Mathematical Sciences — *8–12 Sep 2025*

Summer School: Higher Structures: Recent developments and applications — **Hamburg**
University of Hamburg — *1–5 Sep 2025*

Quantum Field Theory and Topological Phases via Homotopy Theory and Operator Algebras, *MPIM Bonn and CMSA, Harvard* — **Bonn**, *30 Jun – 11 Jul 2025*

Topological Symmetries and Defects in Quantum Field Theory — **Les Diablerets**
SwissMAP Research Station, 2025 SCGCS School — *15–20 Jun 2025*

Higher structures, moduli spaces and integrability — **Hamburg**
University of Hamburg, opening conference of the CRC 1624 — *31 Mar – 2 Apr 2025*

Scottish Talbot On Algebra and Topology — **Glenmore**
Cairngorm Lodge — *23–27 Sep 2024*

IHES Summer School – Symmetries and Anomalies: a modern take **Paris**
Institut des Hautes Études Scientifiques *24 Jun – 5 Jul 2024*

Categorical Symmetries in Quantum Field Theory **Edinburgh**
International Centre for Mathematical Sciences *10–21 Jun 2024*

Conference: Higher Structures in Functorial Field Theory **Regensburg**
University of Regensburg *14–18 Aug 2023*

Summer School: Symplectic Singularities and Supersymmetric QFT **Amiens**
LAMFA, Université de Picardie Jules Verne *10–14 Jul 2023*

TQFTs and their connections to representation theory and mathematical physics, *Hausdorff Center for Mathematics* **Bonn**
19–23 Jun 2023

Winter School in Mathematical Physics **Les Diablerets**
SwissMAP Research Station *8–13 Jan 2023*

Conference: Gauged maps, Vortices and their moduli **Les Diablerets**
SwissMAP Research Station *21–26 Aug 2022*

Further Education and Volunteering

Iliad Intensive **London**
LISA *7 Sep – 2 Oct 2026*
One-month full-time course on mathematical and conceptual foundations of AI safety.

Member of the CdE e.V. **Oberwesel, Bad Hersfeld, Windischleuba**
Participation in various academies *Since 2015*
Attended university-level courses on topics such as quantum algorithms, deep learning, Fermi problems, algebraic topology, English linguistics, inclusive teaching at universities, international law, and designing board games.
Since 2022: leading roles in developing and running systems aimed at fostering inclusivity and safer spaces, and at helping participants with mental health and interpersonal issues.

Additional Skills

Certified Mental Health First Aider **Online**
MHFA Training *26 Feb – 21 Mar 2025*
Programme information available online in German and English

Languages

German	First Language
English	Fluent (TOEFL iBT score of 116/120)